MTH 377/577 Convex Optimization
Problem set 2

1. Let $A = \{(x, y) \in R \times R^2 | x \in R, y \in R^2; y = (x, x^2)\}$ and $B = \{(x, y) \in R \times R^2 | x \in R, y \in R^2; y = (x, \frac{x}{2})\}$. Write down the partial sum of A and B. Is it convex?
   Ans. Partial sum of A, B is $P = \{(x, 2x, x^2 + \frac{x}{2}) \in R^3 | x \in R\}$. Yes, it is convex. (show on your own. hint: let $(x, 2x, x^2 + \frac{x}{2}), (z, 2z, z^2 + \frac{z}{2}) \in P$, pick any $\theta \in [0, 1] \ldots$)

2. Let $f : R \to R$. Define $f(x) = \begin{cases} x^{\frac{1}{3}} & x \geq 0 \\ -x^{\frac{1}{2}} & x < 0 \end{cases}$ Is f convex? Is it quasi-convex? Ans. f is not convex, but it is quasiconvex. Every sub-level set is convex (write down some sub-level sets to demonstrate: $[-1, 1]$ when $\alpha = 1$,$[-8, 8]$ for $\alpha = 2$ etc.).

3. Consider a non-empty, closed, bounded set C. Suppose I can construct a weak separating hyperplane at every $x \in C$ that is on its boundary and not a point in the interior of C. Is C convex?
   Ans. Yes. Here is a simplified indicative proof by contradiction. For the purpose of this question, you can support your explanation diagrammatically with the correct expressions for the separating hyperplane and the point b.
   Proof: Let C be a non-convex, non-empty, closed bounded set that has a weak separating hyperplane at every boundary point. Let $x_1, x_2 \in C$ such that $b = \theta x_1 + (1-\theta)x_2 \notin C$ for some $\theta \in [0, 1]$ and x_0 is a boundary point closest to b. By assumption, $\exists$ weak separating hyperplane (h, β) such that $h^T x \geq \beta \geq h^T b$ for all $x \in C$ and $h^T x_0 = \beta$. Note that for $\theta \in [0, 1]$

$$h^T \theta x_1 \geq \theta\beta \geq h^T \theta b$$
$$h^T (1-\theta)x_2 \geq (1-\theta)\beta \geq h^T (1-\theta)b$$

   Adding the above two, we get

$$h^T [\theta x_1 + (1-\theta)x_2] \geq \beta \geq h^T b$$

Since $b = \theta x_1 + (1-\theta)x_2$, the above holds with equality. Note that the above argument holds for all boundary points of C. Therefore, b lies on every weak separating hyperplane for C i.e. is a point of intersection for all weak separating hyperplanes. This is not possible: Case 1: if C is a line passing through b then $b \in C$ and C is convex. Case 2: if the hyperplanes form a cone with origin at b then C is either not bounded above or below.

4. Let $\succeq$ be a strict binary relation defined over the set $\{a, b, c, d, e\} \in R^5$. Suppose $\succeq$ is reflexive and symmetric, but is not acyclic. Is $\succeq$ an ordering? Why/why not?
   Ans. No, $\succeq$ is not an ordering: suppose $a \succeq b, b \succeq c, c \succeq a$ but not $a \succeq c$. Then there is no order over a, b, c.

5. Let $f_1, f_2, f_3, \ldots f_n$ be convex functions. Is $f(x) = \max\{f_1(x), f_2(x), \ldots, f_n(x)\}$ convex?
   Ans. Yes. try to show on your own.

6. Let $g(\lambda, \upsilon)$ be the standard Lagrangian dual function defined as $g(\lambda, \upsilon) = \inf_{x \in D} L(x, \lambda, \upsilon)$ where D is the domain for x, for some objective function f_0. Suppose f_0 is non-convex. Are L, g convex?
   Ans. By construction L is affine (refer Lecture 10), therefore it is convex and concave both. g maps the pointwise infimums of L. Since L is concave, g is concave.

7. What is the dual of the cone $K = \{\theta x | x = (1, 1, 1, 1, \ldots, 1), x \in R^n_+, \theta \geq 0\}$. Ans. $K* = \{y \in R^n | y^T x \geq 0\} = \{y \in R^n | \sum_{i=1,2,\ldots,n} y_i = 0\}$

8. Let $f : R^3 \to R^3$. Let $g(x) = x_3 f(\frac{x}{x_3})$, where $x \in R^3$. Is g a convex function? Ans. Yes. Use the definition of convex functions to show the same. Also, note that g is a perspective function for f.

9. Let $B = \{\alpha x \in R_n | x \in C, \alpha \in R^+\}$ where $C = \{(1, 3), (3, 1)\}$. Is B a cone? Is it convex? Write down the convex hull of B. Let $\succeq_{B^*}$ be a generalized inequality defined on B^*, the convex hull of B. With the help of an example, show that $\succeq_{B^*}$ is a partial order over R^n.
   Ans. Let $x = (1, 3), y = (3, 1)$. $x - y \notin B^*$ and $y - x \notin B^*$.